

1. Purpose of the Program:

- a. The spaghetti.f program is a sorting algorithm designed to read in and sort a list of numbers using bubble sort; then print them in ascending order. Non-negative numbers are replaced with zero before displaying the final output.

2. How the Original Program Works:

- a. Reads an integer N (the number of values to be entered)
- b. Reads N floating point numbers into an array
- c. Sorts the numbers using the Bubble Sort algorithm
- d. If $N > 10$ it jumps to label 20, skipping some of the program
- e. Replaces negative numbers with 0 but includes another unnecessary GOTO statement.
- f. Prints the sorted list.

3. Issues and Limitations

- a. Overuse of GOTO and unpredictable nested GOTO statements.
- b. Incorrectly implemented BubbleSort.
- c. The GOTO 10 could cause infinite loops.
- d. Some logical issues with 'if' statements.
- e. .GT. and < used in the same program - hard to read.